

RUDRAKSHA SINGH CHAUHAN

MERN Stack Developer | Java & DSA Enthusiast

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EDUCATION

Bundelkhand Institute of Engineering and Technology | Jhansi

B.Tech, Information Technology

CGPA: 8.4

Aug 2023 - Jun 2027

PROJECTS

KrishiSakhi – AI-Powered Farming Assistant | [GitHub](#) [Live Demo](#)

Jul 2025 - Aug 2025

Express.js, MongoDB, Gemini AI, ElevenLabs, Twilio

- Built an AI-powered farming assistant delivering personalized crop recommendations, farming guidance, and agricultural insights using **Gemini AI**.
- Developed RESTful APIs with **Express.js** and **MongoDB** for farmer profiling, activity tracking, and recommendation management.
- Implemented crop disease detection and intelligent pesticide recommendation workflows using image-based analysis and AI-powered decision support.
- Integrated multilingual voice and text interactions using **ElevenLabs** and **Twilio**, enabling farmers to access assistance through natural language communication.

StudyMate – AI-Powered PDF Study Assistant | [GitHub](#) [Live Demo](#)

Jun 2026 - Jul 2026

Next.js 16.2, React 19, Node.js, Express.js, Transformers, Mistral AI, Tailwind CSS

- Built a full-stack AI study assistant using **Next.js 16.2 (App Router, Turbopack)** and **Express.js**, enabling users to upload PDFs and receive AI-generated answers grounded in document content.
- Implemented a PDF ingestion pipeline using **pdf-parse** and generated semantic embeddings with **@xenova/transformers** for similarity-based retrieval (**RAG**).
- Integrated the **Mistral LLM API** for context-aware, source-grounded question answering over uploaded documents.
- Designed RESTful API routes with file-upload handling (**Multer**) and structured logging (**Morgan**).
- Styled a responsive UI with **Tailwind CSS** and deployed on **Vercel**.

StayScape – Property Rental Marketplace | [GitHub](#) [Live Demo](#)

Nov 2024 - Jan 2025

Node.js, Express.js, MongoDB, EJS, Passport.js, Cloudinary

- Engineered a full-stack property rental platform using **Node.js**, **Express.js**, **EJS**, and **MongoDB** within an MVC architecture for property listing and user management.
- Implemented secure user authentication and role-based access control using **Passport.js**, session handling, and **bcrypt** password hashing.
- Developed RESTful CRUD APIs and integrated **Cloudinary** with **Multer** for seamless image uploads, cloud storage, and media optimization.
- Designed scalable MongoDB schemas with referenced data models and centralized async error handling for robust backend validation.

SKILLS

Programming Languages: C, Java, Python, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Redux, Tailwind CSS, Bootstrap, HTML5, CSS3, EJS

Backend: Node.js, Express.js, REST APIs, Mongoose, JWT, Passport.js, OAuth 2.0

AI & Real-Time: Gemini AI, Mistral AI, Transformers, ElevenLabs, WebSockets, Cloudinary

Databases: MySQL, MongoDB, Firebase, Redis

Tools & Platforms: Git, GitHub, Docker, Render, Vercel, VS Code, Postman, Nodemailer

Coursework: Data Structures and Algorithms, Operating Systems, OOP, DBMS, Computer Networks

AWARDS & ACHIEVEMENTS

- Winner, Smart India Hackathon (SIH) 2025** – Secured 1st place in a national-level innovation competition by developing KrishiSakhi, an AI-powered farming assistant featuring crop recommendations, disease detection, and multilingual farmer support.
- Competitive Programming (LeetCode)** – Solved 500+ algorithmic problems and achieved a highest contest rating of 1568, demonstrating advanced proficiency in complex data structure manipulation, dynamic programming, and efficient graph traversal.